

Proposal for Master Thesis: Training-Free Instruction-Following Editing of 3D Human Motion

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1 Motivation and Problem

The production of 3D human animations for virtual reality, games, and film relies heavily on large libraries of unannotated Motion Capture (MoCap) and video-reconstructed clips. When a director or animator wants to refine a specific motion the current pipeline offers no efficient middle ground. Either a technical animator re-rigs the motion by hand, or a supervised machine learning model for editing must be retrained from scratch on thousands of labelled examples. Both approaches are expensive in time, expertise, and data.

Natural language is the most intuitive interface for iterative motion authoring. It requires no domain expertise, it scales to arbitrary instructions, and it matches how creators already communicate with each other. But the current state of the art consists of either supervised editing methods that deliver high semantic quality but fail on any instruction outside their training distribution, which are very small due to data scarcity of (motion, instruction, edit) triplets. Or training-free methods that are robust to distribution shift but limited to low-level kinematic constraints or prompt-level token swaps that cannot handle open-ended, relative instructions on unannotated source clips.

The goal of this thesis is a method that takes any unannotated source motion $\mathbf{x}_0 \in \mathbb{R}^{T \times D}$ together with a free-form natural language instruction e —for instance, “*raise the right arm more*”—and produces an edited motion $\hat{\mathbf{x}}_0$ that faithfully realises the instruction on the targeted body parts and time window while leaving all other joints and frames exactly as they were. Crucially, the method must operate entirely at inference time: no training data, no source text annotation, no model fine-tuning, and no per-sample optimisation are allowed. In the field of image generation and editing a method that satisfies these constraints already exists: LEDITS++ (Brack, Friedrich, Kornmeier, et al., 2024). This thesis proposes a full motion-domain adaptation of LEDITS++, which requires overcoming the unique challenges of extracting anatomically and temporally meaningful masks from the much richer spatiotemporal attention structure of a motion Diffusion Transformer.

Two research questions stand out. The primary question asks whether LEDITS++-style semantic guidance can enable training-free instruction-following editing of 3D human motion. The secondary question asks whether the cross-attention maps of a text-conditioned motion Diffusion Transformer exhibit body-part- and temporally-grounded structure analogous to the spatial structure that LEDITS++ exploits in image diffusion models. The answer to the secondary question determines which masking strategy is used: if cross-attention maps show sufficient anatomical structure, an implicit attention-derived mask is used; otherwise an LLM-derived explicit joint-group mask serves as a fallback.

2 Related Work

Supervised motion editing. is the dominant paradigm, which treats editing as conditional generation conditioned on a source motion and an edit instruction, trained on large collections of (source, instruction, target) triplets. Athanasiou et al. (2024) pioneered this setting with MotionFix/TMED, training on 6,730 labelled triplets; it achieves strong in-distribution quality but degrades sharply on out-of-distribution instructions. SimMotionEdit (Li et al., 2025) and PartMotionEdit follow the same paradigm: both are fully supervised and fail predictably on instructions not seen during training. MotionLab (Guo, Hu, et al., 2025) upgrades the backbone to rectified flows and unifies several motion tasks under a single model, but remains fully supervised and provides no inversion mechanism. The common point of failure for these models is their dependence on a fixed training vocabulary, which extrapolates poorly.

Training-free motion editing. Avoiding labelled data comes at a cost in expressiveness. Karunratanakul et al. (2024) propose DNO, which inverts a source motion via DDIM and then optimises the latent noise to satisfy kinematic constraints; it handles geometric objectives precisely but has no mechanism for semantic natural language instructions. Chen et al. (2025) and Hong et al. (2025) manipulate attention at the prompt-token level, enabling some degree of stylistic control, but both require a source text description of the motion and cannot process relative edit instructions applied to an unannotated clip. In aggregate, training-free methods cover kinematics well but leave semantic instruction-following entirely unaddressed.

Image editing predecessors. LEDITS++ (Brack, Friedrich, Kornmeier, et al., 2024) is the direct image-domain predecessor this thesis adapts to the motion-domain. It combines three independently valuable contributions: edit-friendly DDPM inversion (Huberman-Spiegelglas et al., 2024), which achieves a perfect reconstruction guarantee without requiring a source text prompt; SEGA guidance (Brack, Friedrich, Hintersdorf, et al., 2023), which steers denoising along semantically meaningful directions; and implicit spatial masking derived from cross-attention maps, which localises edits to the regions of an image that are semantically relevant to the edit instruction. The architecture is agnostic to the specific diffusion backbone, which makes domain transfer tractable in principle.

The transfer to motion is, however, non-trivial. LEDITS++ operates on 2D spatial cross-attention maps of size $16 \times 16 = 256$ entries. Motion requires a spatiotemporal map over J joints and F frames; at $J = 22$ body joints and $F = 196$ frames (the maximum HumanML3D (Guo, Zou, et al., 2022) sequence length) this amounts to $22 \times 196 = 4,312$ entries—approximately $17\times$ richer—and must simultaneously encode body-part specificity and temporal locality from a single attention extraction.

Summary. Supervised methods achieve high quality but fail on out-of-distribution instructions; training-free methods handle kinematic constraints but cannot process semantic instructions. This thesis aims to bridge the gap by adapting LEDITS++ to motion, which requires overcoming the challenge of extracting anatomically and temporally meaningful masks from a much richer spatiotemporal attention structure.

3 Solution

The proposed method is a full motion-domain adaptation of LEDITS++ (Brack, Friedrich, Kornmeier, et al., 2024). It runs entirely at inference time on a pre-trained cross-attention motion Diffusion Transformer and requires no training data, fine-tuning, or gradient computation through the model during editing. The pipeline has three sequential stages.

Backbone. A Diffusion Transformer (DiT) is trained on HumanML3D (Guo, Zou, et al., 2022) (263-dimensional joint representation, 14,616 sequences) using a DDPM training objective and sampled with `sde-dpm-solver++`. This specific combination is not incidental: LEDITS++ inversion is derived from the `sde-dpm-solver++` recurrence equations and requires a model that produces DDPM-style noise estimates $\hat{\epsilon}_\theta$. Flow-matching backbones, which output velocity predictions v_θ , are incompatible with this inversion and would forfeit the perfect reconstruction guarantee. Text conditioning is handled through dedicated cross-attention layers—not joint attention—so that cross-attention maps $A^t \in \mathbb{R}^{|c| \times J \times F}$ over the joint-frame space are directly extractable. SMPL joint representation with $J = 22$ joints is used.

Stage 1: edit-friendly inversion. Given a source motion $\mathbf{x}_0$, LEDITS++ inversion (Huberman-Spiegelglas et al., 2024) is derived for the `dpm-solver++` sampler, producing a reconstruction sequence $\mathbf{x}_1, \dots, \mathbf{x}_N$ with a perfect reconstruction guarantee and without any text conditioning. Unlike Null-Text Inversion, no per-timestep optimisation is required. During inversion, the cross-attention maps $A^{t,\ell} \in \mathbb{R}^{|c| \times J \times F}$ are stored across denoising steps and transformer layers for use in the next stage.

Stage 2: spatiotemporal implicit masking. This stage constitutes the core novel contribution. Two complementary binary masks are constructed in $(J \times F)$ space. The semantic mask M^1 is obtained by averaging the stored cross-attention maps over the edit instruction tokens, over denoising steps and over all transformer layers, then thresholding at the λ -th percentile: entry $(j, f) = 1$ if joint j at frame f is semantically relevant to the instruction. The noise-estimate mask M^2 is obtained by computing the element-wise magnitude of the guidance vector $\psi = \epsilon_\theta(\mathbf{x}^t, c_{\text{edit}}) - \epsilon_\theta(\mathbf{x}^t, \emptyset)$ and thresholding likewise; it captures fine-grained motion boundaries in joint-frame space. The final mask $M = M^1 \cap M^2$ (element-wise product) is simultaneously semantically grounded and fine-grained.

Stage 3: SEGA-Guided denoising with multi-edit composition. Denoising from $\mathbf{x}_N$ proceeds using the edit instruction c_{edit} . At each timestep t , the masked guidance is applied (Huberman-Spiegelglas et al., 2024):

$$\hat{\epsilon}_0(x_t, c_e) = \hat{\epsilon}_0(x_t, \emptyset) + \sum_{i=0}^K s_e \cdot M_i \cdot [\epsilon_\theta(x_t, c_e) - \epsilon_\theta(x_t, \emptyset)] \quad (1)$$

where per-instruction masks $M_i = M_i^1 \cap M_i^2$ prevent cross-edit interference when multiple simultaneous instructions K are provided. After each denoising step, frames identified as temporally unedited by M (all-zero columns in M) are replaced with the exact source motion value at the corresponding noisy timestep. This guarantees zero drift in unedited frames—a strict constraint that LEDITS++ does not need for images but is essential for physical plausibility in motion.

4 Evaluation

MotionFix benchmark. The primary evaluation is made on the full MotionFix test set (Athanasios et al., 2024), without source prompts and without training on any MotionFix data. The metrics are R@1, R@2, R@3 (Recall at k (Petrovich et al., 2023)), FID (motion quality), and MPJPE on unedited joints (source preservation). Comparisons are drawn against the available data of training-free baselines DNO (Karunratanakul et al., 2024), MotionCLR (Chen et al., 2025), and SALAD (Hong et al., 2025), and against the supervised upper bounds TMED (Athanasios et al., 2024) and SimMotionEdit (Li et al., 2025). The expected outcome is that

the proposed method outperforms all training-free baselines on semantic edit metrics while closing, but not eliminating, the gap to supervised methods on in-distribution instructions.

Out-of-distribution generalisation. 150 edit instructions of types absent from MotionFix training are constructed manually: cross-cultural gesture styles (e.g., a formal Japanese bow), VR-specific actions (e.g., reaching upward for a virtual object), and compound multi-part instructions (e.g., “*use the opposite arm*”). Each instruction is applied to five HumanML3D (Guo, Zou, et al., 2022) test motions, giving 750 edited sequences. The same metrics are recorded. This experiment is designed to isolate the generalisation advantage conferred by the training-free approach.

Ablations. A structured ablation study disentangles the contribution of each design decision: removing the mask entirely versus using only M^1 or only M^2 versus the intersection $M^1 \cap M^2$; comparing the implicit attention-derived M^1 against the explicit LLM-derived fallback; comparing edit-friendly DDPM inversion (Huberman-Spiegelglas et al., 2024) against standard DDIM inversion; sweeping the guidance scale $s_e \in \{1, 2.5, 5, 7.5, 10\}$; and evaluating single-edit versus multi-edit composition with and without hard frame inpainting.

5 Extension Point

The following directions extend the core method naturally. They are presented in roughly increasing order of implementation effort and serve either as thesis extensions if the primary timeline permits or as directions for future work.

Full-body editing with hand articulation. Primary evaluation uses only the 22 body joints of SMPL, excluding the 30 hand joints, to maintain comparability with MotionFix baselines (Athanasίου et al., 2024). The spatiotemporal mask architecture is agnostic to J and requires no modification to operate over the full $J = 52$ joint set, which would enable fine-grained editing of hand gestures—for example, “*open the hand slowly*” or “*point with the index finger.*”

Longer sequences and temporal transitions. HumanML3D (Guo, Zou, et al., 2022) caps sequences at 196 frames (roughly 10 seconds). Real production clips are longer and involve natural transitions between action phases. Applying the method via a sliding window and evaluating on transition-editing tasks—for instance, “*make the transition to the jump more explosive*”—would characterise the temporal boundary effects of the hard-inpainting strategy at larger scales and reveal any artefacts introduced at window boundaries.

Cross-modal instruction sources. The method currently accepts text as the edit signal. Because the DiT’s cross-attention layers use the same CLIP embedding space that processes text tokens, visual references—such as a short reference video clip or a skeletal keyframe sketch—could be projected into that space and used as edit signals without modifying the denoising pipeline. This would open the method to workflows where creators prefer to demonstrate rather than describe a desired motion change.

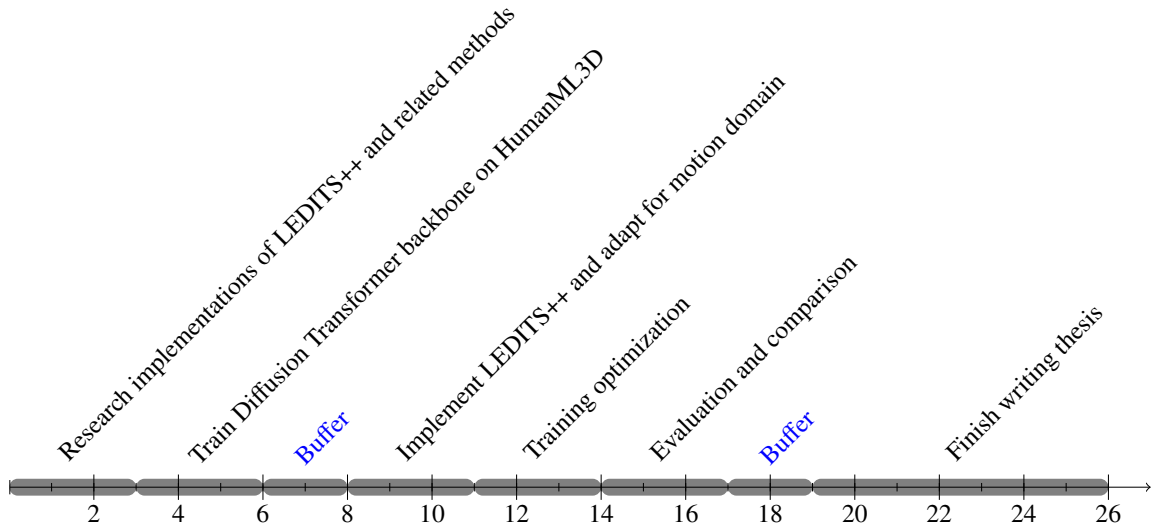


Figure 1: Timeline in weeks since the registration.

6 Timeline

Figure 1 sketches the allocation of the 26 weeks between thesis registration and submission. The first phase is dedicated to a thorough study of the literature, with particular attention to existing implementations of LEDITS++ (Brack, Friedrich, Kornmeier, et al., 2024) and the training pipelines of related motion diffusion models. This groundwork is essential before any implementation begins.

The second phase covers backbone development: training the Diffusion Transformer on HumanML3D (Guo, Zou, et al., 2022) and validating that the resulting model satisfies the two prerequisites for LEDITS++ inversion—a DDPM noise-prediction objective and extractable cross-attention maps over joint-frame space. Depending on the results of early training runs, this phase may involve experimenting with different model sizes, architectural variants, or training strategies to obtain a backbone whose cross-attention maps exhibit sufficient anatomical structure.

With a working backbone in hand, the third phase turns to the core contribution: implementing the edit-friendly inversion, the spatiotemporal implicit masking, and the SEGA-guided denoising pipeline, followed by the full evaluation protocol described in Section 4. The final weeks are reserved for consolidating the results and completing the written thesis.

If time permits, the extension points outlined in Section 5 can be explored in parallel with the main timeline.

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